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Detection of CITES-listed *hoodia* species and extracts thereof in herbal supplements using LC-HRMS/MS: validation study and qPCR method comparison

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ABSTRACT

This study describes the validation of a approach to detect in herbal supplements regulated *Hoodia* species or their extracts, targeting the oxypregnane steroidal glycoside P57AS3, using liquid chromatography coupled to tandem mass spectrometry with a high-resolution acute mass detection (LC-HRMS/MS). The method was validated by assessing the selectivity and specificity by analysing other related Apocynaceae species, particularly *Stapeliinae* subtribe members, and various plants commonly found in dietary supplements promoting weight loss. The limit of detection (LOD), equal to 8 µg P57AS3 per gram powder, matrix effects, and the ratio of false positives (0%) were determined. Next, the analytical sensitivity of the LC-HRMS/MS methodology was compared to a recently developed quantitative PCR (qPCR) method by comparing the results obtained from 20 commercial herbal supplements, often originating from rogue internet pharmacies, that either claimed or were reported to contain *Hoodia* sp. material or extracts thereof. The study results highlight the complementary nature of LC-HRMS/MS and qPCR. While qPCR provides high molecular sensitivity for powdered material, LC-HRMS/MS delivers chemical confirmation and detection of metabolites when DNA is absent or degraded. The integration of both techniques creates a comprehensive detection framework, enhancing accuracy, reliability, and confidence in the identification of regulated plant material across different and complex herbal matrices and mixtures thereof.

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Introduction

The global market for weight loss dietary supplements is expected to triple by 2030 (Crawford et al. 2024). These supplements, purchased to attain or maintain leaner body weight, are thought to act either by reducing hunger and therefore food intake, reducing the absorption of nutrients, stimulating the body to burn fat, or even having laxative effects. Common ingredients in these supplements include dietary fibre, caffeine, and minerals with or without herbal components. Often these botanical components originate from extracts or dried herbal material from *Camellia sinensis* (tea plant), *Garcinia cambogia*, *Citrus*

aurantium (bitter orange), *Senna alexandrina*, *Ilex paraguariensis* (Yerba mate), artichoke, and several spices such as cayenne pepper and turmeric. In the past, dietary supplements containing *Ephedra* sp. as a source for ephedrine or isomers were very popular as weight loss enhancers (Diepvens et al. 2007; Mund and Frishman 2013). However, as the United States and Europe banned the use of plants in this genus due to undesirable side effects (Nelson 2004; European Commission, Commission Regulation 2015/403 2015), the market turned to other plant species, including *Hoodia gordonii* (Brendler 2020). This plant is a slow-growing spiny succulent from the subtribe *Stapeliinae*, a member of the Apocynaceae family,

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and grows naturally in the south of Africa (Namibia, Botswana, South Africa). There, it has been used by local San bushmen to fight hunger and thirst during long hunting trips and at times of famine (van Heerden 2008). From 1997 to 2007, the plant and its extracts were studied by different multinationals (e.g., Phytopharm, Pfizer, and Unilever) for their proclaimed weight loss effects. In 2005, the plant species and the rest of the genus were listed in Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), meaning that trade is permitted in wild and artificially propagated material with appropriate permits. However, a couple of years later, the *Hoodia* bubble started to burst as clinical studies showed no significant weight loss between the usage of the extract and a placebo, and more importantly, several adverse effects were reported (Le Nevé et al. 2010; Madgula et al. 2010; Blom et al. 2011; Smith and Krygsman 2014). Moreover, at that time, many of the dietary products sold claiming to contain *H. gordonii* turned out to be adulterated with more abundant species, such as *Opuntia*, that did not necessarily contain the relevant fraction of steroidal glycosides, such as the purported weight loss molecule P57AS3 (CAS number 384329-61-7) (Avula et al. 2008; Joshi et al. 2009). Although many companies have ceased their research and investment, herbal supplements claiming to contain *H. gordonii* can still be frequently found on e-commerce platforms, even though the plant and extracts thereof are considered unauthorised food ingredients within the EU (European commission, implementing regulation 2018/1133 n.d) or are actively banned (Agence nationale de sécurité du médicament et des produits de santé (ANSM) 2020). Nevertheless, a study assessing the prevalence of warnings in the Rapid Alert System for Food and Feed (RASFF) from the European Union demonstrated that in 2017, almost 30 notifications were made for herbal supplements containing this plant species (Koncz et al. 2021). Even in 2023 and 2024, notifications were made of *H. gordonii*-containing supplements that resulted in serious adverse effects. Needless to say, proper detection of *H. gordonii* and genus members remains relevant for law enforcement agencies due to their CITES

status and continuous occurrence and promotion of *Hoodia gordonii* supplements on the illegal market, either through rogue internet pharmacies or available in bulk quantities through different e-commerce platforms. Previously published detection methodologies, developed more than a decade ago, have taken two main approaches: targeting the *Hoodia* biomarker P57AS3 by screening for its sodium adduct (m/z 901.94) using liquid chromatography coupled to mass spectrometry (LC-MS) (Avula et al. 2017), or utilising molecular technologies such as PCR followed by sequencing of specific barcoding regions (e.g., internal transcribed spacer 1 (ITS1) or *psbA-trnH*) to detect this plant species (Joshi et al. 2009; Gathier et al. 2013). While the LC-MS methodology was validated for *Hoodia* powder, little to no validation was performed for herbal mixtures, despite many supplements containing them (Avula et al. 2008, 2017). To our knowledge, no validation was conducted for the described qPCR methodology, although validation is often a prerequisite for such analyses when seizures are involved. Recently, multidimensional fingerprinting combined with chemometrics has been employed to distinguish *H. gordonii* from other components in herbal matrices (Ranjan et al. 2023). However, the data processing associated with this analytical approach is time-consuming and unsuitable for the relatively rapid analysis often required by regulatory agencies. Furthermore, closely related family members were not assessed in this study, leaving the specificity of this approach questionable. More recently, a novel qPCR methodology employing an internal probe and the ITS2 region as an amplification target has been developed and validated for detecting *Hoodia* sp. in herbal mixtures (Fraiture et al. 2026). In this study, we developed a targeted liquid chromatography coupled to high-resolution tandem mass spectrometry (LC-HRMS/MS) approach for detecting P57AS3 in *Hoodia* products and compared it with this validated qPCR method (Fraiture et al. 2026). The development of the LC-HRMS/MS approach was necessary to achieve a selective and specific methodology. The method was validated according to ISO17025 and International Conference on Harmonisation (ICH) requirements (ISO/IEC 17025 2017; ICH 2023), and following the

guidelines established by the United Nations for Testing of Illicit Drugs in Seized Materials and Biological Specimens (United Nations Office on Drugs and Crime (UNODC) 2009). Here, we evaluated the LC-HRMS/MS method's selectivity and specificity by testing against other Apocynaceae species, particularly members of the Stapeliinae sub-tribe, and common plant species found in dietary supplements. We also determined the limit of detection (LOD), matrix effects, and the ratio of false positives and false negatives (UNODC 2009). Both methodologies were then applied to a pilot study of 20 commercial dietary supplements claiming or reported to contain *Hoodia* material or extracts. This approach aimed to assess the performance and complementarity of molecular (qPCR) and chemical (LC-HRMS/MS) techniques across different product types, including both raw powder and processed extract matrices, to establish a comprehensive analytical framework for verification of *Hoodia* in dietary supplements.

Materials and methods

Solvents and reference standards for chemical analysis

Mass spectrometry (MS)-grade acetonitrile (purity 99.8%), methanol (purity > 99.9%), water, and formic acid (purity > 99%) were purchased from Thermo Fisher Scientific (Waltham, MA, USA). The reference standard for P57AS3 (CAS 384329-61-7) was custom synthesised by Prof. Dr. Biao Yu (Zhang et al. 2012; Wang et al. 2023), and the commercial version of this molecule, used for validation studies, was purchased from Biosynth (Staad, Switzerland).

Herbal reference material

The herbal reference material, listed in the supplemental data (Table S1), was obtained from multiple sources: botanical gardens, the American Herbal Pharmacopoeia (AHP), certified reference material from the Joint Research Centre (JRC), wild-harvested specimens, and commercially available certified reference material. The herbal reference material was, when applicable, cut into small pieces with a surgical blade, and held in an incubator at 65 °C until the material was deemed dry. This dry material was then ground to a

powder by an analytical mill (IKA-Werke GmbH, Germany). This powder was stored at −20 °C until further analysis.

Sample set

The herbal supplements were either bought online from e-commerce sites, indicating the presence of a warehouse within Europe, or were collected by regulatory agencies and claimed to contain *Hoodia* sp., or were thought to contain *Hoodia* sp. based on prior chemometric analysis (Ranjan et al. 2023). The supplemental data (Table S2) provides more details on the 20 different samples.

Sample preparation for chemical analysis

For liquid chromatography coupled to tandem mass spectrometry (LC-HRMS/MS), 250 mg of dried material was weighed and extracted by adding 4 mL of methanol. This mixture was sonicated for 15 min, followed by centrifugation at 8 000 × g for 2 min before filtration (0.22 µm PTFE filter) of the supernatants and subsequent injection into the system. All herbal supplements were screened for the presence of P57AS3 based on previously described sample preparation (Cohen et al. 2024). Briefly, 5 capsules or tablets were weighed, opened, and their contents were mixed and triturated in a mortar and pestle. Next, about 250 mg of this mixture was weighed and resuspended in 4 mL of methanol prior to sonication for 15 minutes. Next, the mixture was centrifuged for 10 minutes at 8 000 × g prior to injection into the LC-MS/MS system.

LC-HRMS/MS analysis

Instrumental settings: High-resolution accurate mass tandem MS (HRMS/MS) analyses were carried out on a Thermo Scientific™ Vanquish™ ultra-high-performance liquid chromatography (UHPLC) system coupled to a Q Exactive™ focus orbitrap mass spectrometer (Thermo Fisher Scientific, Bremen, Germany). The optimised LC methodology, employing an Acquity UPLC BEH C18 column (100 mm × 2.1 mm, 1.7 µm particle size) (Waters, Milford, MA, USA), was as follows: isocratic elution for 1 min at 50% mobile phase B (0.1% formic acid in acetonitrile), 50% mobile phase A (0.1% formic acid in water), at a

constant flow rate of 0.4 mL/min and a column temperature of 45 °C, followed by a steep increase to 95% B at 4.0 min and held isocratic at 95% B for 1.5 min prior a final re-equilibration step of 1.5 min with 50% mobile phase A, resulting in a total run time of 7 min. To our knowledge, this is currently the fastest methodology described (Avula et al. 2008, 2017).

The high-resolution accurate mass analysis was performed on a Q-Exactive Focus Orbitrap MS (Thermo Fisher Scientific, Bremen, Germany), equipped with a heated electrospray ionisation (ESI) source operating in positive ion mode. The optimised tune method was set as follows: the flow rate of sheath gas (nitrogen, purity $\geq 99.99\%$) and auxiliary gas (nitrogen, purity $\geq 99.99\%$) was set at 30 and 10 arbitrary units, respectively; the temperature of the capillary was set at 320 °C; the voltage of the spray was 3.5 kV; and the S-lens RF level was set at 60 V. The MS scan data were acquired at a resolving power of 70,000 (at m/z 200). The MS/MS data, with a resolving power of 17,500, were obtained with the use of parallel reaction monitoring (PRM) scanning mode for the identification of metabolites. The collision energy for the targeted precursor ion-triggered collision-induced dissociation (CID) was adjusted to 40%. All data were acquired using the Thermo Xcalibur 4.0 software, and processed by Trace-Finder 5.1 software (Thermo Fisher Scientific, Bremen, Germany). A typical spectrum is shown in Figure 1.

DNA analysis

For each sample, the DNA extraction was performed as previously described using NucleoSpin® Food Kit (MACHEREY-NAGEL,

Germany) (Fraiture et al. 2026). DNA concentration was measured by fluorometry using Qubit 3.0 Fluorometer (Thermo Fisher, Belgium), and the DNA purity was evaluated based on the A260/A280 and A260/A230 ratios provided by spectrophotometry using Nanodrop® 2000 (Thermo Fisher, Belgium). The *Hoodia*-specific qPCR method was performed as recently described (Fraiture et al. 2026), while the plant-specific method was performed as described by Laube *et al.* (Laube et al. 2010). The used oligonucleotide sequences are listed in Table S3 of the supplemental data. For each commercial sample, 25 ng of DNA was tested in duplicate. Next, for each commercial sample, potential PCR inhibition was evaluated using the plant-specific qPCR method. A total of 25 ng of DNA and its 5-fold dilution was analysed in duplicate. The measured and theoretically expected quantification cycle (C_q) values were compared. PCR inhibition was considered when the deviation between the observed and expected ΔC_q was more than 1.

Results and discussion

Choice of biomarker, development, and optimisation of the LC-HRMS/MS screening methodology

To determine the optimal chromatographic methodology for the detection of the oxypregnane steroidal glycoside P57AS3, several C18 columns and MS-compatible solvents were tested. This compound was selected from among several available biomarkers due to its commercial

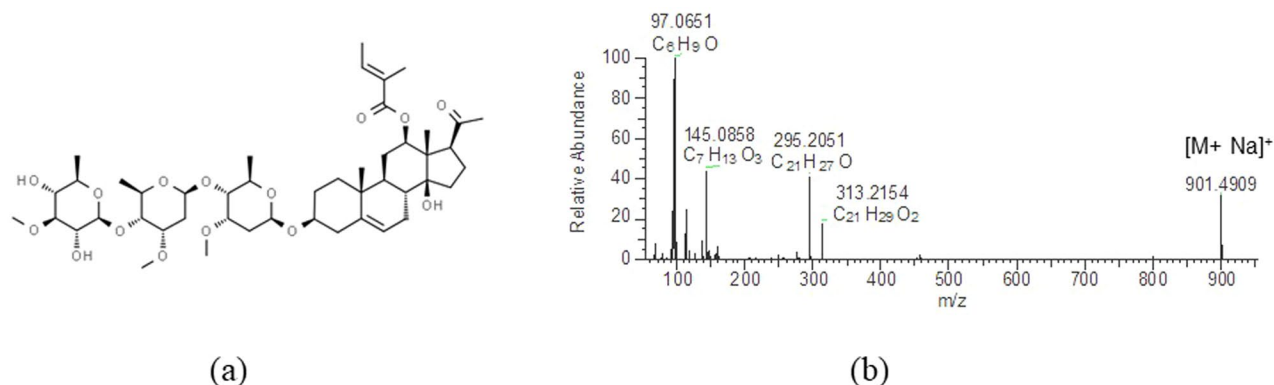


Figure 1. Chemical structure of P57AS3 (CAS number 384329-61-7) and the LC-HRMS/MS spectrum.

availability in Europe, as other potential biomarkers proved difficult to obtain.

Validation of the LC-HRMS/MS screening methodology

The parameters tested to validate the screening method included selectivity and specificity. This was achieved by checking the absence of a component with a similar retention time (tolerated error of 0.1 min.) and similar MS/MS spectrum (see material and methods) as the reference standard, and assessing methanolic extracts of the different related plant species and other plant species often present in proclaimed weight loss enhancing herbal supplements (see Table 1 and

Figure 2). The presence of component P57AS3 was confirmed only when all of the following criteria were strictly met: (1) the retention time difference was ≤ 0.1 min compared to the reference standard; (2) the m/z of the precursor ion matched that of the reference standard within an error tolerance of 5 ppm; and (3) the MS/MS spectrum contained the remnant precursor ion (m/z 901.492 at 50% relative intensity) along with the three most intense fragment ions observed for the reference standard (m/z 295.205 at 50%, m/z 145.056 at 50%, and m/z 97.065 at 100%). Compliance with the following relative intensity tolerances was mandatory: for relative intensities between 40 and 60%, a maximum relative error of 25% was permitted; for relative intensities

Table 1. Summary of the analytical results obtained with LC-HRMS/MS analysis and qPCR analysis (Fraiture et al. 2026) for closely related plant species and other plant species often found or claimed to be present in weight loss herbal supplements.

Related plant species (Apocynaceae)				
Subfamily	Tribe	Species	Presence of P57AS3	Presence of Hoodia DNA
Asclepiadoideae	Ceropegieae-subtribe stapeliinae	<i>Hoodia macrantha</i>	+	+
		<i>Hoodia gordonii</i> (1)	+	+
		<i>Hoodia gordonii</i> (2)	+	+
		<i>Hoodia gordonii</i> (3)	+	+
		<i>Hoodia juttae</i>	+	+
		<i>Hoodia parviflora</i> (1)	+	+
		<i>Hoodia parviflora</i> (2)	+	+
		<i>Hoodia pilifera</i>	+	+
		<i>Hoodia pedicellate</i>	+	+
		<i>Hoodia ruschii</i>	+	+
		<i>Caralluma fimbriata</i>	–	–
		<i>Caralluma europaea</i>	–	–
		<i>Caralluma hesperidum</i>	–	–
		<i>Caralluma joannis</i>	–	–
		<i>Ceropegia woodii</i>	–	–
		<i>Huernia keniensis</i>	–	–
		<i>Huernia rubra</i>	–	–
		<i>Orbea variegata</i>	–	–
		<i>Stapelia flavirostris/grandiflora</i>	–	–
		<i>Stapelia romotriche revoluta</i>	–	–
	Marsdenieae	<i>Dischidia nummularia</i>	–	–
		<i>Dischidia ruscifolia</i>	–	–
		<i>Dischidia sp.</i>	–	–
		<i>Hoya multiflora</i>	–	–
Rauvolfioideae	Vinceae	<i>Rauvolfia vomitoria</i>	–	–
Apocynoideae	Nerieae	<i>Strophanthus amboensis/petersianus</i>	–	–

Plant species often used in herbal slimming supplements				
Family	Common name	Scientific name	Presence of P57AS3	Presence of Hoodia DNA
Apiaceae	Fennell	<i>Foeniculum vulgare</i>	–	–
Aquifoliaceae	Yerba maté	<i>Ilex paraguariensis</i>	–	–
Asteraceae	Artichoke	<i>Cynara scolymus</i>	–	–
Cactaceae	Opuntia	<i>Opuntia ficus indica</i>	–	–
Clusiaceae	Garcinia	<i>Garcinia cambogia/gummi-gutta</i>	–	–
Fabaceae	Senna	<i>Senna alexandrina</i>	–	–
Rutaceae	Bitter orange	<i>Citrus aurantium</i>	–	–
Sapindaceae	Guarana	<i>Paullinia cupana</i>	–	–
Solanaceae	Cayenne pepper	<i>Capsicum annuum</i>	–	–
Theaceae	Green tea	<i>Camellia sinensis</i>	–	–

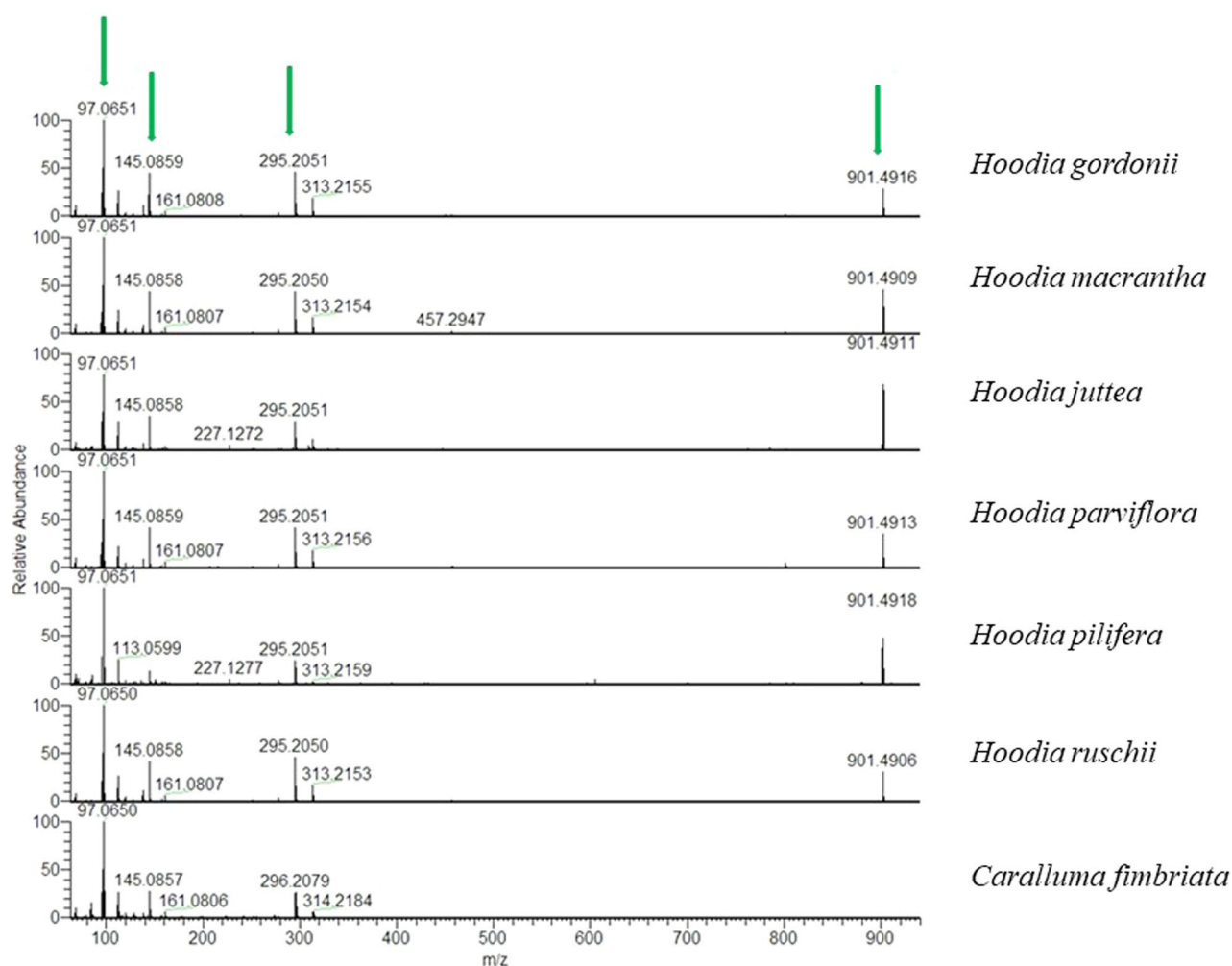


Figure 2. The HRMS/MS spectra obtained for a precursor ion with an m/z of 901.492 eluting at 3.8 minutes.

above 60%, a maximum relative error of 10% was acceptable. All criteria had to be simultaneously satisfied for positive identification.

As shown in Table 1 and depicted in Figure 2, P57AS3 was only found in the tested *Hoodia* species. Similar findings were reported in a previous study by Avula et al. (2008), but they contrast with the results from Gathier et al. (2013). These researchers found that P57AS3 was not detected in *H. parviflora* and *H. triebnerii*, but it was present in species from the same subtribe, such as *Orbea hardyi* and *Stapelia hirsute*. Although *H. triebnerii* was not examined here, analysis of the methanolic extracts of two *H. parviflora* plants confirmed the presence of P57AS3. Likewise, *Orbea hardyi* and *Stapelia hirsute* were not analysed in this study; however, other members of these genera tested negative for the targeted compound.

The different targeted HRMS chromatograms for *Hoodia* sp., closely related species (*Orbea variegata*, *Stapelia grandiflora*, and *Ceropegia rupicola*), and various other plant species are shown in Figure 3 and demonstrate the absence of a peak with an m/z of 901.492 at 3.8 minutes, except for the methanolic extract of *Caralluma fimbriata* (see Figure 2). However, the HRMS/MS spectrum of this species lacked the presence of one of the fragmentation ions (m/z 295.205), indicating that the molecule present is likely not P57AS3. Nonetheless, *Caralluma* contains several steroidal saponins that might structurally be very similar to P57AS3 (Choucry et al. 2021). Based on the above data, it can be stated that this methodology resulted in 0% false positives, which is essential for screening methods used by regulatory agencies and in court cases (UNODC 2009). In addition to the selectivity and

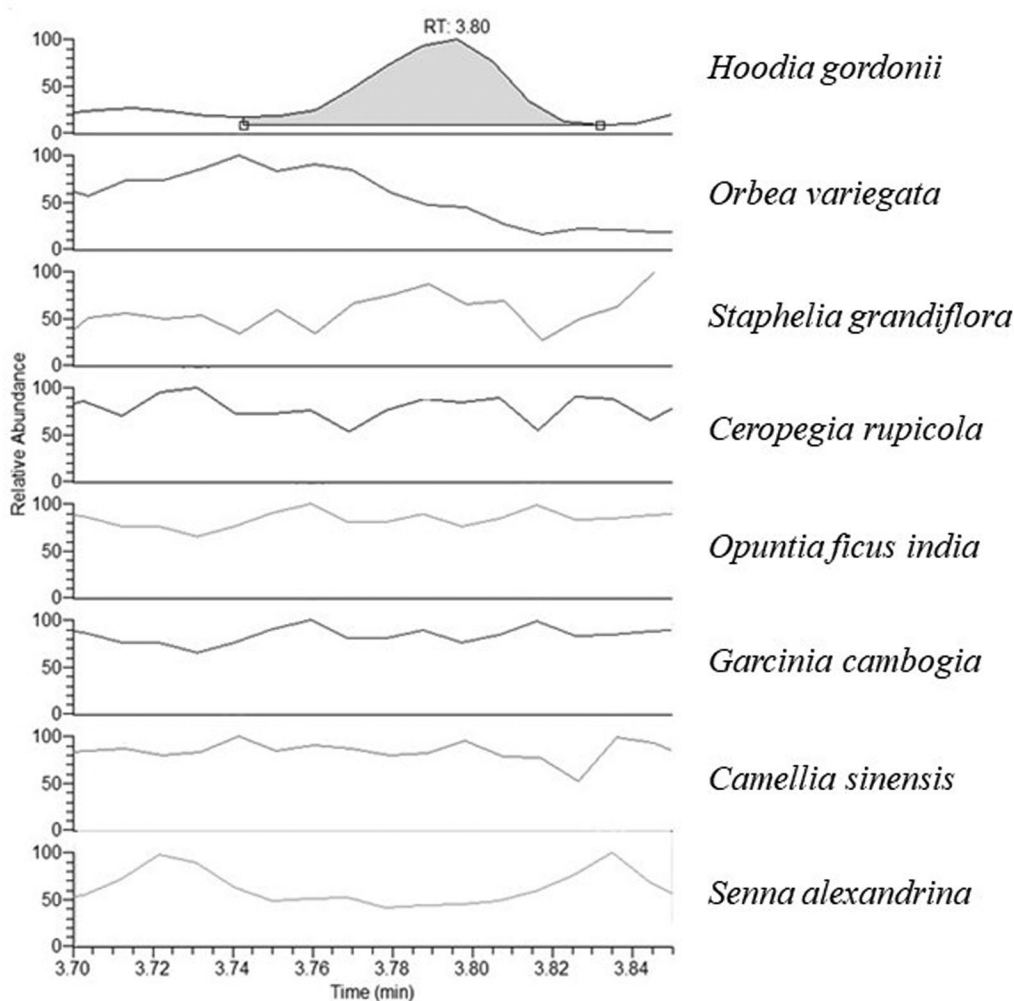


Figure 3. The LC chromatogram indicating the occurrence of a precursor ion with an m/z of 901.492 eluting at 3.8 minutes in methanolic extracts of *H. gordonii* and its absence in methanolic extracts of other plant species.

specificity, concurrent with the guidelines, the limit of detection (LOD) and the ratio of false positives and false negatives were determined. The LOD of P57AS3 was determined by serial dilution of the reference standard in different methanolic extracts of different plant species other than *Hoodia* species (*Orbea variegata*, *Stapelia grandiflora*, *Ceropegia rupicola*, *Caralluma fimbriata*, *Opuntia ficus indica*, *Garcinia cambogia*, *Camellia sinensis*, and *Senna alexandrina*). The lowest concentration where the molecule could still be detected in all matrices was taken as the LOD and corresponded to $0.5 \mu\text{g/mL}$, equivalent to $8 \mu\text{g}$ per gram powder. Based on scarce occurrence data, the lowest amount of P57AS3 reported in *Hoodia* sp. corresponded to approximately $50 \mu\text{g}$ per gram of material (Avula et al. 2006), resulting in an extract with a

concentration of P57AS3 of approximately $3 \mu\text{g/mL}$. This indicates that the developed methodology is sensitive enough and fit for purpose. Next, methanolic extracts of ten plant species often present in herbal supplements were spiked at 1 time the LOD of P57AS3, and in all cases, the targeted molecule was found, as requested by the guideline (UNODC 2009).

Analysis of commercial samples by targeted LC-HRMS/MS and comparison with qPCR assays

Before the analysis of the different herbal supplements, the labels of the products were screened for the presence of a batch number and an expiration date, which are mandatory in the EU (European commission, directive 2011/91/EU and Regulation No 1169/2011 n.d-a, n.d-b). As shown

in Table 2, 15% of the samples of this pilot study (3/20) lacked either a lot number, expiration date, or both. Half of the samples had an unauthorised health claim present on their label. Nine samples (sample 1–9) mentioned the presence of *Hoodia* powder or root, nine other samples (sample 10–18) mentioned the presence of *Hoodia* extract, and two samples were included in this study as the presence of *Hoodia* sp. was suspected based on previous data employing multidimensional chromatographic fingerprinting combined with chemometrics (Ranjan et al. 2023).

All samples were analysed using the developed targeted LC-HRMS/MS method (Table 2). These samples were also analysed by DNA-based methods, using both plant-specific and *Hoodia*-specific qPCR assays (Table 2). The presence of PCR inhibitors was observed in samples 8 and 16 (Supplemental data, Table S4). In total, 35% of the samples (7/20) were positive for the presence of P57AS3, while 25% of the samples (5/20) were positive according to the qPCR targeting *Hoodia* DNA. As can be observed in Table 2, similar qPCR and MS results were obtained for those samples that listed the presence of *Hoodia* powder or root, except for sample 4. For this sample, qPCR indicated the presence of *Hoodia* sp., although with a relatively high C_q value (see Table S4), suggesting a low amount of target

DNA in the sample. In contrast, the LC-HRMS/MS data indicated no detectable P57AS3. This may reflect the inherently higher specificity of qPCR methodology compared to the chemical approach. Moreover, it is also likely that the P57AS3 concentrations might also vary depending on the plant and the different growth conditions. For this reason, when analysing suspected *Hoodia* powder, the qPCR methodology might be favoured over the chemical approach, or ideally, both methodologies should be employed to ensure comprehensive detection. However, when analysing *Hoodia* extracts (samples 10–18), no *Hoodia*-specific DNA was detected in several of those samples. However, in samples 12, 16, and 18, P57AS3 was detected by LC-HRMS/MS, indicating that *Hoodia* extracts were present, but DNA had likely been degraded or removed during the extraction process. This is consistent with common P57AS3 extraction methods, which use harsh conditions employing different solvents, such as methanol and heptane, known to cause DNA denaturation and fragmentation (Herskovits et al., 1961). Moreover, more than half of the extract samples (5/9) contained detectable plant DNA, but no *Hoodia*-specific DNA was found. This could reflect the presence of plant species in the samples, as four of these five samples listed additional plant material on the label (see Table

Table 2. Information provided on the label compared with the outcome of the LC-HRMS/MS and qPCR analysis.

n°	Health claim present on the label	Labelled amount of <i>Hoodia</i> per capsule or tablet	Lot number and expiration data mentioned	Presence of P57AS3	Presence of plant DNA	Presence of <i>Hoodia</i> sp. DNA
1	–	200mg <i>Hoodia gordonii</i> powder (aerial parts)	Y	+	+	+
2	–	200mg <i>Hoodia gordonii</i> powder (aerial parts)	Y	+	+	+
3	–	350mg <i>Hoodia gordonii</i> powder	Y	+	+	+
4	–	1000mg <i>Hoodia gordonii</i> powder	Y	–	+	+
5	Hunger control-herbal supplement	Amount not specified	Y	–	–	–
6	Metabolism booster-herbal supplement	Amount not specified	Y	–	–	–
7	Metabolism support	133.3mg <i>Hoodia gordonii</i> powder (aerial parts)	Y	+	–	+
8	Weight loss supplement- appetite suppressant	Amount not specified	Y	–	– ^a	–
9	Appetite-suppressant	Amount not specified	Y	–	–	–
10	–	50:1 extract	N	–	–	–
11	–	100:1 extract	N	–	–	–
12	Appetite reducer	<i>Hoodia parviflora</i> dry extract	Y	+	–	–
13	–	60mg <i>Hoodia gordonii</i> extract (20:1)	No lot number	–	+	–
14	–	400mg <i>Hoodia</i> extract (1:30)	Y	–	+	–
15	–	400mg <i>Hoodia</i> extract (1:30)	Y	–	+	–
16	Diet Support for Energy Metabolism	<i>Hoodia gordonii</i> extract (1:20)	Y	+	– ^a	–
17	–	20:1 extract	Y	–	–	–
18	Appetite suppressant	<i>Hoodia gordonii</i> extract (12:1)	Y	+	+	–
19	Controls appetite- promotes weight loss	Not mentioned on the label	Y	–	–	–
20	Appetite control	Not mentioned on the label	Y	–	–	–

^aSamples showing a significant inhibition for PCR amplification.

S2). Primer bias favouring actin amplification is unlikely, as demonstrated by sample 7, where *Hoodia*-specific DNA was detected with high sensitivity, confirming that the qPCR method reliably identifies *Hoodia* when present. Sample 13 was the only sample that did not list the presence of other plant material, suggesting that labelling errors or the occurrence of undeclared plant species are not that uncommon, certainly when dealing with supplement products being available through rogue internet pharmacies (Ekar and Kreft 2019; Ichim 2019; Gafner et al. 2023). Surprisingly, the two samples (19 and 20) that were reported as positive for *Hoodia gordonii* by multidimensional chromatographic fingerprinting combined with chemometrics (Ranjan et al. 2023), contain neither *Hoodia*-specific DNA nor P57AS3. One possible explanation is that these samples represent *Hoodia* extracts derived from plants that contained only very low amounts of P57AS3, which consequently may have remained below the detection limit of our methodology. However, further analysis for other *Hoodia* glycosides using extracted ion chromatography as described by Avula et al. 2017, no corresponding m/z were found, suggesting that these samples are likely false positives for *Hoodia gordonii*.

Taken together, these findings suggest an analytical strategy for an unknown sample. If a sample is positive for P57AS3, it can be assumed to

contain *Hoodia* sp. Conversely, a negative result warrants further analysis by qPCR to check for the presence of *Hoodia* DNA before drawing any conclusion. If the qPCR yields negative results, either no *Hoodia* DNA is present in the samples (e.g. extracts), or significant PCR inhibition took place; consequently, more in-depth analysis might be required to verify the presence/absence of other *Hoodia*-related molecules, as by Avula et al. 2017, which might become quite lengthy and cumbersome. However, reference standards, required to confirm their identity according to ICH and the guidelines put forward by the UNODC, are not commercially available. Therefore, it can be concluded that if both employed methodologies are negative, *Hoodia* remains undetected in the sample and can be reported as such. Based on the abovementioned data from market samples, a decision tree is depicted in Figure 4.

It must be noted that the aforementioned approaches can only be used to identify plants belonging to the *Hoodia* genus and do not allow for differentiation between species within the genus. This distinction could be significant because, in 2018, the European Food Safety Agency (EFSA) authorised only the dried aerial parts of *Hoodia parviflora* for market use. The applicant, Israel-based Desert Labs, was permitted by EFSA, under the novel food regulation, to market herbal

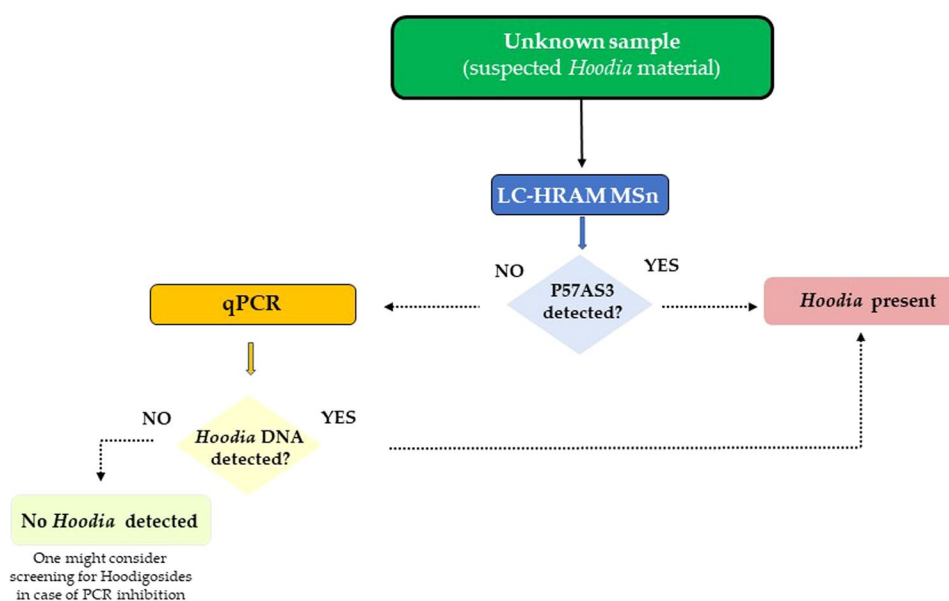


Figure 4. Decision tree for *Hoodia* sp. detection in unknown samples.

supplements containing this plant, provided they target only adults and limit the maximum daily intake to 9.4 mg of dried aerial powder per day (European Commission, implementing regulation 2018/1133). However, this amount is considerably lower than the quantities often claimed in herbal supplements (see Table 2) and at least 15 times less than the amount used by the applicant or shown to be effective for weight loss in consumer trials (Landor et al. 2015). As a result, at least to our knowledge, the applicant has not marketed herbal supplements with this limited amount of *Hoodia parviflora* in the EU. Furthermore, most online food supplements or raw materials, whether bulk or extracts, claim to contain *Hoodia gordonii* rather than *Hoodia parviflora*. Nonetheless, future molecular techniques, such as DNA sequencing, could be employed to distinguish *H. parviflora* from other *Hoodia* species, as genetic differences at certain loci have been reported in prior studies (Joshi et al. 2009). However, since the abovementioned study only included three *Hoodia* species, caution is advised. The selectivity and specificity of such molecular methods need to be carefully validated using various plants from the same species and other species within the *Hoodia* genus (Gathier et al. 2013). Despite these current limitations in species-level differentiation, the genus-level identification methods presented in this study provide a valuable foundation for regulatory screening.

Conclusions

This study established and validated a targeted LC-HRMS/MS methodology for the detection of the oxypregnane steroidal glycoside P57AS3 in *Hoodia* sp. material and extracts, providing a complementary approach to existing molecular techniques for identifying this CITES-listed plant in dietary supplements. The validation demonstrated high selectivity and specificity, with comprehensive assessment against related *Apocynaceae* species and common dietary supplement ingredients. Although the current methods enable genus-level identification rather than species-level differentiation, they provide a valuable foundation for regulatory screening as the entire genus is listed under CITES-II except for small amounts (9.4 mg) of dried aerial parts of *H. parviflora* which are

allowed if properly labelled. However, the majority of the samples available online or those analysed in this study list the occurrence of *H. gordonii*, which still remains an unauthorised food.

Subsequent application of both LC-HRMS/MS and qPCR methods to 20 commercial products revealed distinct performance profiles across different sample types. Both techniques yielded consistent results for products containing *Hoodia* powder; however, only LC-HRMS/MS successfully detected P57AS3 in extract-based supplements where DNA was likely degraded or absent. This finding underscores the critical importance of method selection based on sample processing state: molecular approaches perform optimally with minimally processed materials, whereas chemical methods provide robust detection across diverse matrices, including highly processed extracts. Taken together, the integration of both chemical and molecular analytical strategies created a comprehensive framework for regulatory enforcement and quality control of products claiming to contain *H. gordonii*. This dual-method approach can be extended to other botanicals where both DNA-based and chemical marker detection are feasible, enhancing confidence in authentication across various processing conditions.

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Author contributions

CRedit: **Celine Vanhee**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft; **Andrea Gobbo**: Formal analysis, Investigation, Methodology, Writing – review & editing; **Pauline Guinot**: Formal analysis,

Investigation, Writing – review & editing; **Marie Willocx**: Formal analysis, Investigation, Writing – review & editing; **Nancy Roosens**: Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – review & editing; **Marie-Alice Fraiture**: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

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Data availability statement

Data is contained within the article or supplementary material.

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